# 7.2 Properties of Parallelograms

Name _____ Period ____ Date _____

Geometry 7.2 (30-minute plan) · Student sheet

**I can…**

- use the sides and angles of a parallelogram to find missing measures.
- use the fact that the diagonals of a parallelogram bisect each other.
- find a missing vertex or the intersection of the diagonals in the coordinate plane.

**Vocabulary** · fill in the missing word

**Parallelogram** — a _____ with both pairs of opposite sides parallel

**Consecutive angles** — angles at the two _____ of one side

**Bisect** — cut into two _____ parts

## Part 1 · Notes: fill in the blanks

Opposite sides of a parallelogram are _____.

Opposite angles of a parallelogram are _____.

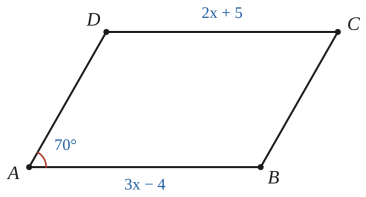
Consecutive angles of a parallelogram are _____ (they add to _____°).

The diagonals of a parallelogram _____ each other.

## Part 2 · Examples (work them with the class)

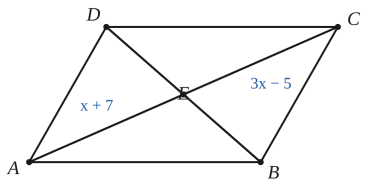
### Example 1

ABCD is a parallelogram. AB = 3x − 4, DC = 2x + 5, and m∠A = 70°. Find x, AB, m∠B, and m∠C.




### Example 2

In parallelogram ABCD, the diagonals meet at E. AE = x + 7 and EC = 3x − 5. Find x and AC.



## Part 3 · Partner practice

### Partner practice

In parallelogram QRST, the diagonals meet at E. QE = 2y + 1 and ES = 4y − 9. Find y and QS.